

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/386331928>

Artificial Intelligence (AI) and Health Communication Policy in Nigeria: Challenges and Prospects

Article · December 2024

DOI: 10.52589/JARMS-ARD2E2R4

CITATIONS

13

READS

198

1 author:



[Nonye Benedeth Ezeaka](#)

Chukwuemeka Odumegwu Ojukwu University

32 PUBLICATIONS 137 CITATIONS

SEE PROFILE



ARTIFICIAL INTELLIGENCE (AI) AND HEALTH COMMUNICATION POLICY IN NIGERIA: CHALLENGES AND PROSPECTS

Nonye Benedeth Ezeaka (Ph.D.)

Department of Mass Communication, Chukwuemeka Odumegwu Ojukwu University,
Igbariam Campus.

Email: nb.ezeaka@coou.edu.ng, ezeakanonye79@gmail.com; Tel.: 08037711374

Cite this article:

Ezeaka, N. B. (2024),
Artificial Intelligence (AI) and
Health Communication Policy
in Nigeria: Challenges and
Prospects. Journal of
Advanced Research and
Multidisciplinary Studies 4(4),
141-149. DOI:
10.52589/JARMS-
ARD2E2R4

Manuscript History

Received: 16 Sep 2024

Accepted: 26 Nov 2024

Published: 2 Dec 2024

Copyright © 2024 The Author(s).

This is an Open Access article
distributed under the terms of
Creative Commons Attribution-
NonCommercial-NoDerivatives
4.0 International (CC BY-NC-ND
4.0), which permits anyone to
share, use, reproduce and
redistribute in any medium,
provided the original author and
source are credited.

ABSTRACT: *Advances in Artificial Intelligence (AI) are reshaping health communication worldwide, a key aspect of enhancing public health. This study analyzed current health communication policies to pinpoint areas where AI-driven approaches are yet to be addressed. In Nigeria, integrating AI into health communication policies presents both opportunities and hurdles. The study examined how AI can boost health communication initiatives in Nigeria, particularly through AI-powered tools that enhance the spread of health messages, patient education, and public awareness campaigns. Additionally, the study highlighted challenges in implementing AI technologies, including limited infrastructure, inadequate data management, and a shortage of skilled professionals. The paper also presented the potential of Artificial Intelligence (AI) to transform health communication. It suggested policy guidelines to support its integration while reducing risks. Additionally, it recommended policy changes, training, and cooperation among stakeholders to leverage AI's power in health communication. In Nigeria, the outlook for AI in health communication is optimistic, assuming focused investments in technology, training, and regulatory reforms.*

KEYWORDS: Artificial intelligence, Health communication, Policy, Nigeria, Challenges, Prospects.



INTRODUCTION

AI is changing industries by improving how we get information and services. In health communication, AI can help doctors make decisions, improve patient outcomes, and make healthcare systems more efficient. Nigeria faces health challenges like high rates of infectious diseases, poor healthcare infrastructure, and growing non-communicable diseases. Using AI in health communication could be both an opportunity and a challenge.

In Nigeria, health communication policy plays a vital role in combating health inequalities and ensuring that up-to-date, accurate health information is disseminated to the public. However, there are several obstacles to overcome, including a lack of reliable infrastructure, limited data literacy, and restricted access to technology. Ethical issues, data privacy concerns, and the need for regulatory frameworks further complicate the use of artificial intelligence in this area. It is widely believed that the future lies in the information age, where a nation's success hinges on its ability to handle information effectively (Enemuo, Ezeanyi & Ezeaka 2019).

Despite difficulties, AI holds great promise for improving health communication. AI can analyze large amounts of data to spot trends in public health, tailor health messages to particular groups, and predict disease outbreaks. Additionally, AI-powered tools can enable real-time communication between patients and healthcare professionals, fostering health literacy and engagement. This study investigates the challenges and prospects of integrating AI into Nigeria's health communication policy. It examines AI adoption in Nigeria's healthcare industry, identifies barriers to effective AI-driven health communication, and suggests solutions and policy suggestions for leveraging AI's potential for better health communication.

Nigeria's people have different backgrounds, speak different languages, and have different beliefs about health, so it's important to communicate health information in different ways. Traditional communication methods may not work as well when technology is changing quickly and not everyone has access to digital devices. AI can be used to improve the way health information is made, spread, and read by people.

The COVID-19 outbreak has highlighted the urgent need for clear health communication, especially during emergencies. In Nigeria, the spread of false information on social media created public confusion and mistrust about prevention measures and vaccines. AI tools like NLP and machine learning algorithms can be used to find and stop this misinformation, helping health officials give the public the right information at the right time. Also, because many Nigerians use smartphones to find health information, the rise of mobile technology in Nigeria gives a unique chance to use AI to spread health information.

Using AI in health communication brings challenges. Healthcare workers, policymakers, and the public need to be trained to use these technologies effectively. The AI systems need to be transparent, culturally appropriate, and ethical to build trust and give everyone equal access to health information. In addition, data privacy and security must be addressed since health data is very sensitive.

AI integration in health communication in Nigeria offers great potential for better health outcomes. However, it also brings challenges that need careful attention. This paper explores the current health communication landscape in Nigeria and assesses the role of AI technology. It suggests actionable strategies to improve health communication policy. By doing so, it aims



to guide policymakers, healthcare providers, and researchers on how AI can enhance health communication in Nigeria, making it more effective and accessible for all.

Objectives of the study

The objectives of the study are to:

1. Assess the current state of AI adoption in Nigeria's healthcare sector; and to
2. Identify the challenges and prospects of AI-driven health communication in Nigeria.

Statement of the Problem

The health communication landscape in Nigeria is fraught with challenges that hinder the effective dissemination and understanding of critical health information. Despite advancements in technology and a rising mobile phone penetration rate, there remains a substantial gap in health literacy among the population (Akinwunmi & Akinyemi, 2020). This gap is further exacerbated by the spread of misinformation, especially during public health crises like the COVID-19 pandemic, leading to harmful health choices (Ogunyemi et al., 2021). Many individuals struggle to differentiate between accurate health information and false claims, resulting in detrimental health outcomes (Nwankwo, 2022).

Traditional health communication strategies often fall short in addressing the diverse cultural and linguistic backgrounds of Nigeria's populace. A significant lack of tailored communication approaches that resonate with various communities leads to inadequate engagement and compliance with health directives (Adebayo et al., 2019). Compounding this issue is the limited availability of health communication resources and trained professionals who can effectively utilize modern technologies (Ndukwe et al., 2020).

While artificial intelligence (AI) holds promise for transforming health communication by enhancing information delivery, personalizing health advice, and countering misinformation (Wathore & Rao, 2021), its integration into Nigeria's health communication strategies remains inadequate and inconsistent. There is an urgent need to explore how AI technologies can be effectively employed to improve health communication, identify existing barriers to their adoption, and formulate policies that promote equitable access to accurate health information. Communication is central to every awareness campaign (Ezebuonyi & Ezeaka, 2015).

Overall, this study seeks to address the pressing issue of ineffective health communication in Nigeria by investigating the role of AI innovations in enhancing health literacy, combating misinformation, and informing policy development. By focusing on these critical areas, the research aims to contribute to more effective and inclusive health communication strategies that can significantly improve public health outcomes in Nigeria.



THEORETICAL FRAMEWORK: HEALTH BELIEF MODEL (HBM)

The Health Belief Model (HBM) serves as the primary theoretical framework for this study. Developed by Rosenstock in the 1970s, the HBM is a psychological model that seeks to explain and predict health behaviors by focusing on individual beliefs and attitudes about health issues. The model identifies several key constructs that influence health behavior: perceived susceptibility, perceived severity, perceived benefits, and perceived barriers (Rosenstock, 1974).

The relevance of the Health Belief Model to the study on health communication in Nigeria lies in its ability to provide insights into how individuals interpret and respond to health information—especially in the context of misinformation and the use of artificial intelligence (AI) technologies. Here are several ways in which the HBM applies to this study:

1. **Understanding Health Literacy:** By focusing on perceived susceptibility and severity, the HBM allows for an examination of how individuals assess their vulnerability to health issues (e.g., infectious diseases) and their awareness of potential health outcomes. This is particularly pertinent in Nigeria, where varying levels of health literacy can lead to differing responses to health messages.
2. **Identifying Barriers:** The model's emphasis on perceived barriers can help identify obstacles that Nigerian populations face in accessing and acting upon health information. Barriers may include cultural misconceptions, language differences, or mistrust of health messaging from authorities. With insights derived from the HBM, AI can be tailored to address these barriers effectively.
3. **Enhancing AI-Driven Interventions:** As the study explores the role of AI in health communication, the HBM can inform the design of AI tools that increase perceived benefits and reduce perceived barriers. For example, AI systems could be programmed to present information in culturally relevant ways that resonate with the target audience's beliefs and experiences, thereby enhancing the likelihood of behavior change.
4. **Promoting Behavior Change:** The HBM can guide the development of strategies that encourage individuals to adopt healthier behaviors, particularly in the face of health crises like the COVID-19 pandemic. By understanding the factors influencing people's beliefs, health communicators can craft messages that inspire confidence in the benefits of following health guidelines communicated through AI channels.

The Health Belief Model serves as a foundational framework for this study, enabling a nuanced understanding of how individual beliefs about health shape responses to health communication in Nigeria. By applying the HBM, the research can effectively explore how artificial intelligence can enhance public health outcomes through tailored and effective communication strategies.



METHODOLOGY

This research uses a qualitative method to explore the complexities of integrating artificial intelligence (AI) into healthcare communication policies in Nigeria. This approach enables a thorough analysis of the challenges and opportunities associated with AI adoption in healthcare. The study analyzed important policy papers, reports, and materials on AI in healthcare to identify areas of concern and room for improvement.

Current State of AI Adoption in Nigeria's Healthcare Sector

In Nigeria, the use of artificial intelligence (AI) in healthcare is becoming more common. AI is being used to improve health results, make services better, and solve long-standing problems like not being able to get care, having a lot of diseases, and not having enough resources. This review of the literature gives an overview of how AI is currently being used in the Nigerian healthcare system, focusing on important applications, barriers to implementation, and opportunities for growth.

Current Applications of AI in Healthcare

AI is being utilized in some domains of the Nigerian healthcare sector, each contributing to improved patient care and operational efficiency:

- **Diagnostic Tools:** AI-powered diagnostic tools have emerged as a key application, helping healthcare professionals to analyze medical images, such as X-rays and CT scans, with greater accuracy. Initiatives like the Deep Learning-based Radiology programs have shown promise in detecting diseases like tuberculosis and cancer, where radiological expertise is often scarce (Adeyemi et al., 2020).
- **Health Data Management:** AI is being leveraged to organize and interpret large volumes of healthcare data, facilitating better decision-making. Tools that utilize natural language processing (NLP) are being used to extract relevant information from electronic health records (EHRs), leading to more efficient patient management and enhanced quality of care (Ogunyemi et al., 2021).
- **Predictive Analytics:** Predictive analytics powered by AI can anticipate disease outbreaks and identify high-risk populations. For instance, models analyzing social media data and epidemiological patterns have been deployed to predict outbreaks of diseases like cholera and malaria, enabling proactive responses (Adebayo et al., 2021).
- **Telemedicine and Virtual Care:** The COVID-19 pandemic accelerated the integration of AI in telemedicine platforms, which are increasingly being adopted for remote consultations and health monitoring. AI chatbots and virtual health assistants have been employed to provide preliminary assessments, schedule appointments, and answer patient queries, thereby widening access to care in remote areas (Soyinka et al., 2022).

Barriers to AI Adoption

Despite potential benefits, several barriers hinder the widespread adoption of AI in Nigeria's healthcare sector:



- **Infrastructure Challenges:** Inadequate infrastructure, such as unreliable internet connectivity and limited access to high-quality computing resources, poses a serious challenge to the implementation of AI technologies in healthcare settings (Eze et al., 2021).
- **Data Privacy and Security Concerns:** The collection and processing of health data raise concerns about patient privacy and data security. Issues surrounding data governance and the ethical use of AI algorithms need to be addressed to build trust among stakeholders (Nwafor et al., 2023).
- **Lack of Skilled Workforce:** There is a significant shortage of healthcare professionals with the necessary skills to implement and effectively use AI technologies. Training and capacity-building initiatives are essential to bridge this skills gap (Ibrahim et al., 2022).
- **Regulatory and Policy Frameworks:** The absence of comprehensive regulatory frameworks to govern the use of AI in healthcare may hinder innovation and adoption. Establishing clear policies that address ethical considerations, accountability, and standards is critical for fostering a conducive environment for AI deployment (Okoroiwu et al., 2023).

Opportunities for Growth

Despite the challenges, several opportunities exist for the future growth of AI in Nigeria's healthcare sector:

- **Public-Private Partnerships:** Collaborations between government entities and private organizations can lead to increased investment in AI technologies and infrastructure development. Such partnerships can also support innovative pilot projects and scale successful initiatives (Orji et al., 2021).
- **Local AI Solutions:** Investing in locally developed AI solutions tailored to the specific needs and contexts of Nigerian healthcare can enhance adoption rates. Involvement from local stakeholders and healthcare providers in the development process can ensure relevance and effectiveness (Oluikpe et al., 2022).
- **Investment in Education and Training:** Strengthening educational programs focused on health informatics, data science, and AI can help cultivate a skilled workforce capable of driving AI adoption in health care. This could involve collaborations with universities and professional organizations (Nwachukwu et al., 2022).

The current state of AI adoption in Nigeria's healthcare sector reflects both significant potential and formidable challenges. While advancements in diagnostic tools, data management, predictive analytics, and telemedicine indicate a growing integration of AI technologies, barriers such as infrastructure deficiencies, privacy concerns, and a lack of skilled personnel must be addressed to fully realize the benefits of AI. By fostering public-private partnerships, investing in local solutions, and enhancing education and training, Nigeria can position itself to leverage AI in improving health outcomes and transforming its healthcare landscape.



CONCLUSION

The integration of artificial intelligence (AI) into Nigeria's healthcare sector presents a transformative opportunity to address longstanding challenges in service delivery, disease management, and patient outcomes. This study has explored the current state of AI adoption in Nigeria, highlighting its applications in diagnostic tools, health data management, predictive analytics, and telemedicine. These applications hold the potential to revolutionize healthcare by improving the accuracy of diagnoses, streamlining operations, and expanding access to essential health services, particularly in underserved and remote areas.

However, the study has also identified significant barriers that impede the effective implementation of AI technologies. These include infrastructural deficiencies, concerns around data privacy and security, a shortage of skilled healthcare professionals, and the absence of robust regulatory frameworks to guide the ethical use of AI. Addressing these challenges is critical for fostering an environment conducive to the successful integration of AI in healthcare.

Nevertheless, numerous opportunities exist for fostering growth and expansion of AI in Nigeria's healthcare landscape. Public-private partnerships can facilitate investment and infrastructure development, while locally-driven AI solutions ensure that the technologies are tailored to the specific needs and cultural contexts of Nigerian communities. Furthermore, enhancing education and training programs can equip healthcare professionals with the necessary skills to harness AI capabilities effectively.

Ultimately, this study underscores the importance of a multi-faceted approach that encompasses technological, ethical, and educational dimensions to fully leverage AI's potential in Nigeria's healthcare system. By navigating the obstacles and capitalizing on the opportunities identified, Nigeria can position itself at the forefront of AI-driven healthcare innovation in Africa. Moving forward, stakeholders—including government entities, private sector organizations, health professionals, and educational institutions—must collaborate to establish a cohesive strategy for the sustainable adoption of AI in healthcare, ensuring that it not only enhances health outcomes but also promotes equity and accessibility for all citizens.

Practical Recommendations for AI Adoption in Nigeria's Healthcare Sector

To effectively leverage artificial intelligence (AI) in Nigeria's healthcare sector, several practical recommendations can be implemented, considering the unique socio-economic, cultural, and infrastructural context of the country:

1. Infrastructure Development

Invest in improving internet and telecommunications infrastructure, especially in rural areas, to enable reliable access to AI-driven healthcare applications. Creating AI research and innovation centers in partnership with universities and research institutions to develop localized solutions that address specific healthcare challenges in Nigeria.

2. Capacity Building and Education

Implement training programs focused on AI and health informatics for healthcare professionals to build a skilled workforce adept at using AI tools in clinical settings. Integrating AI and data analytics courses into medical and nursing school curricula to prepare future healthcare professionals for a technology-driven environment.



3. Public-Private Partnerships (PPPs)

Collaborations between government, private companies, and NGOs to fund and scale AI initiatives in healthcare should be encouraged. This can leverage financial resources, technology, and expertise. Creating incentives, such as grants or tax breaks, for startups developing innovative AI solutions tailored to local healthcare needs.

4. Local Content Development

Encourage the development of context-specific AI solutions created by Nigerian tech firms to address local health issues, focusing on languages, cultural practices, and specific health challenges prevalent in different regions. Involving healthcare professionals, community leaders, and patients in the design and implementation of AI tools to ensure they meet the practical needs of users.

5. Funding and Investment

Increase government funding for AI-related health initiatives within national health budgets, recognizing the importance of digital health transformation. Seeking partnerships with international organizations and agencies for funding and technical assistance to implement AI projects in Nigeria's healthcare system.

6. Community Awareness and Engagement

Raise awareness about the benefits of AI in healthcare through community outreach programs, workshops, and media to improve acceptance and understanding. Communication is essential for people to interact effectively (Ezeoke, Ezeaka & Nwodu, 2020). Leveraging community health workers as intermediaries to promote AI health initiatives, demonstrating their value in local settings. Initiating conversations and involving communities in interaction can help achieve the objectives set out (Ezeaka & Ochuba, 2024).

REFERENCES

- Adebayo, A. M., Owoaje, E. T., & Dairo, M. D. (2019). Health communication and public compliance with health policies: Evidence from Nigeria. *Journal of Health Communication*, 24(11), 889-895. <https://doi.org/10.1080/10810730.2019.1655916>
- Adebayo, K., Ojo, J. A., & Akintayo, C. O. (2021). Predicting disease outbreaks using machine learning: A focus on cholera and malaria in Nigeria. *Nigerian Journal of Clinical Practice*, 24(3), 509-516.
- Adeyemi, O., Olayiwola, E., & Akinkunmi, A. (2020). The application of artificial intelligence in radiology: A review of progress in Nigeria. *Journal of Medical Imaging and Radiation Sciences*, 51(2), 145-151.
- Akinwunmi, B., & Akinyemi, J. (2020). Assessing health literacy among Nigerian citizens: A systematic review. *Nigerian Journal of Health Education*, 15(3), 1-14.
- Enemuo, C.I., Ezeanyi, B.C. & Ezeaka, N.B (2019). Extent of Information Communication Technology (ICT) Integration among students in Tertiary Institutions in Anambra State. *International Journal of Education and Research* 7(7), 75- 84.



- Eze, A., Okeke, M. O., & Agunwamba, J. C. (2021). Infrastructure and healthcare transformation in Nigeria: Exploring the inadequacies and opportunities for AI adoption. *International Journal of Health Management and Information*, 6(1), 23-35.
- Ezeaka, N.B., & Ochuba, C.C. (2024). Harnessing AI in Development Communication for Drug Abuse Prevention: A Nigerian Perspective. *Mass Media Review* Vol 6 (1)
- Ezeaka, N.B & Nwodu, E.G (2022). *Communication for Partnership in Development*. In A.N Nwammuo; G.U Nwafor & B.N Ogbonna (eds) Twenty-One Scholars' Viewpoints on Development Communication. Enugu: Rhyce Kerex Publisher
- Ezebuenyi, E.E & Ezeaka, N.B. (2015). Diffusion of Social Change Campaigns in Rural Communities: The Critical Role of Trado-Modern Media Approach. *Mass Media Review* (2) 1 69-81
- Ezeoke, C.B., Ezeaka, N.B., & Nwodu, G.E. (2020). *Understanding Health Communication*. In C.S. Okunna (ed.), *Communication and Media Studies, Multiple Perspective*. Enugu: New Generation Educare Ltd.
- Ibrahim, A. A., Aderibigbe, B., & Olowookere, S. A. (2022). Bridging the skills gap in healthcare AI: A systematic review of training programs in Nigerian institutions. *Health Informatics Journal*, 28(1), 146-162.
- Ndukwe, C. D., Ajuwon, A. J., & Alabi, O. (2020). Challenges and prospects of health communication in Nigeria: The role of information technology. *Journal of Communication in Healthcare*, 13(2), 65-72. <https://doi.org/10.1080/17538068.2020.1792547>
- Nwachukwu, C. C., Oloyede, O. O., & Nduka, F. (2022). Addressing the AI skills gap in Nigeria's healthcare sector: Role of education and training. *Journal of Healthcare Engineering*, 2022, Article ID 9788704.
- Nwafor, C., Eze, O., & Agbo, G. (2023). Data privacy and security in AI healthcare applications: Ethical considerations in the Nigerian context. *Nigerian Journal of Health Policy and Management*, 4(1), 15-28.
- Nwankwo, E. (2022). Misinformation and its effects on health decisions in Nigeria: Policy recommendations. *African Journal of Health Management*, 8(1), 25-31.
- Ogunyemi, A., Igbokwe, I., & Adeyemi, M. (2021). Public health and misinformation: Lessons from the COVID-19 pandemic in Nigeria. *International Journal of Public Health*, 66, 640344. <https://doi.org/10.3389/ijph.2021.640344>
- Oluikpe, P. I., Talabi, F. O., & Idowu, O. (2022). Leveraging local innovations in AI for healthcare: The Nigerian experience. *Global Health Action*, 15(1), 2042176.
- Okoroiwu, H., Ogujiofor, G. A., & Udeh, E. (2023). Regulatory frameworks for artificial intelligence in healthcare: Emerging trends and challenges in Nigeria. *Nigerian Journal of Legislative Studies*, 12(2), 32-45.
- Orji, A. K., Ogbole, O. J., & Ijeoma, E. (2021). Transforming healthcare in Nigeria through public-private partnerships: Opportunities and challenges. *Public Health*, 190, 162-169.
- Soyinka, B., Adeyemo, A., & Folayan, M. (2022). The impact of telemedicine on healthcare delivery during the COVID-19 pandemic in Nigeria: A review. *Journal of Telemedicine and Telecare*, 28(1),
- Rosenstock, I. M. (1974). Historical origins of the Health Belief Model. *Health Education Monographs*, 2(4), 328-335. <https://doi.org/10.1177/109019817400200403>
- Wathore, R., & Rao, G. (2021). Artificial intelligence in health communications: Transformative potential and challenges. *Medical Informatics and Decision Making*, 21(1), 23. <https://doi.org/10.1186/s12911-021-01497-4>